



1.5 Matrices and systems of equations (HL)

Number and algebra · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Matrices can be added, scaled, and multiplied (when dimensions are compatible). For a 2×2 matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $\det(A) = ad - bc$, and if $\det(A) \neq 0$, $A^{-1} = (1/\det A) \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. A system of linear equations $Ax = b$ can be solved as $x = A^{-1}b$ when $\det(A) \neq 0$ (unique solution); if $\det(A) = 0$, the system has either no solution or infinitely many.

Worked example: follow the reasoning

Solve the system $2x + y = 8$, $x - 3y = -3$ using matrices.

Solution: $A = \begin{bmatrix} 2 & 1 \\ 1 & -3 \end{bmatrix}$, $b = (8, -3)$. $\det(A) = 2(-3) - 1(1) = -7$. $A^{-1} = (1/-7) \begin{bmatrix} -3 & -1 \\ -1 & 2 \end{bmatrix}$. $x = A^{-1}b = (1/-7)(-3 \times 8 - 1 \times (-3), -1 \times 8 + 2 \times (-3)) = (1/-7)(-21, -14) = (3, 2)$. So $x = 3$, $y = 2$.

Quick check · multiple choice

For a 2×2 matrix A to have an inverse, what condition must hold?

- A $\det(A) = 0$
- B $\det(A) \neq 0$
- C A must be symmetric
- D A must be diagonal

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A small factory produces two products, X and Y. Each unit of X needs 3 hours of labour and 2 kg of material; each unit of Y needs 2 hours of labour and 4 kg of material. In one week, 130 hours of labour and 180 kg of material are fully used. (a) Write this situation as a matrix equation for the number of units x of X and y of Y produced. (b) Solve the system using matrices to find x and y . [6]

Final answer: _____



Answers and explanation

Question 1 [6]

(a) $\begin{bmatrix} 3 & 2 \\ 2 & 4 \end{bmatrix} (x, y)^t = (130, 180)^t$. (b) $\det = 3 \times 4 - 2 \times 2 = 8$. Inverse $= (1/8) \begin{bmatrix} 4 & -2 \\ -2 & 3 \end{bmatrix}$. $(x, y) = (1/8)(4 \times 130 - 2 \times 180, -2 \times 130 + 3 \times 180) = (1/8)(160, 280) = (20, 35)$. So 20 units of X and 35 units of Y are produced.

Quick check answer

B. A matrix is invertible exactly when its determinant is non-zero.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics AI subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.